

Critical Slowing Down

Bogner Alina
fc66044

Laier Benjamin
fc66045

Deraï Bérénice
fc66034

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Abstract

This project explores critical slowing down in the two-dimensional Ising model near the critical temperature T_c . Using Markov chain Monte Carlo methods, we implement both the Metropolis and Wolff algorithms to generate spin configurations and measure autocorrelation times of magnetization and energy. Magnetic susceptibility is determined with an external field B , and, in the absence of the field, via the fluctuation-dissipation theorem (FDT). The scaling of relaxation times allows estimation of the dynamic critical exponent z for each algorithm. Comparisons between local and cluster updates highlight their relative efficiency at criticality, showing how algorithmic choices affect the sampling of Markov chains in systems exhibiting critical slowing down and field-induced responses.

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1 Introduction

The Ising model is a fundamental model in statistical mechanics used to study phase transitions and critical phenomena. Despite its seemingly simple definition, including binary spin variables and nearest-neighbor interactions, it captures the essential features of spontaneous symmetry breaking and critical behaviour observed in even more complex systems. In two dimensions, the Ising model exhibits a well-known second-order phase transition at a critical temperature T_c , where various physical quantities such as magnetization, susceptibility, and correlation length display power-law divergences characterized by critical exponents.

To study both static and dynamic properties of the Ising model, Monte Carlo simulations based on Markov-chain sampling are widely used. Near the critical point, these algorithms are subject to **critical slowing down**, where the relaxation and autocorrelation time increase rapidly as the correlation length diverges. In practice, this means that successive configurations generated by the simulation become highly correlated, leading to inefficient sampling and large statistical errors. Understanding this is essential for accurate numerical studies close to T_c .

This study investigates the effective dynamics of the 2D Ising model as generated by different Monte Carlo update algorithms: the local Metropolis algorithm, which performs local single-spin updates, and the global Wolff cluster algorithm, which employs non-local cluster flips. Although both algorithms sample from the same equilibrium distribution, they impose different dynamical behaviors on the system, leading to distinct scaling behaviours of the autocorrelation time near the critical point. This scaling is characterized by the **dynamic critical exponent** z , which determines the divergence of time scales according to $\tau \sim \xi^z$ as the correlation length $\xi \rightarrow \infty$ at criticality.

The goals of this report are:

1. To derive and discuss the theoretical background of dynamic critical behaviour and the definition of the dynamic exponent z .
2. To measure autocorrelation times of magnetization and energy for both Metropolis and Wolff dynamics as the temperature approaches T_c .
3. To compare the scaling and efficiency of the two algorithms in the critical region.

Section 2 introduces the theoretical framework, including the Ising model, Monte Carlo dynamics, and the concept of dynamic scaling. Section 3 outlines the simulation methodology and measurement procedures. Section 4 presents the numerical results and analysis, while Section 5 discusses the implications of the findings and compares algorithmic performance. Finally, Section 6 summarizes the main conclusions.

2 Dynamic Critical Exponents and Autocorrelation Times

2.1 The 2D Ising Model

The two-dimensional square-lattice Ising model is a paradigmatic system for studying phase transitions and critical phenomena. Spins $s_i = \pm 1$ occupy the sites of the $L \times L$ lattice with periodic boundary conditions (PBD) and interact with their nearest neighbors. The Hamiltonian for the system is given by:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i, \quad (1)$$

where J is the interaction strength, h is an external magnetic field, and the sum $\langle i,j \rangle$ runs over nearest neighbor pairs.

In zero external field ($h = 0$), the model exhibits a second-order phase transition at a critical temperature T_c , separating a low-temperature ferromagnetic phase (with non-zero spontaneous magnetization) from a high-temperature paramagnetic phase (with zero magnetization). For the 2D Ising model, the critical temperature is known exactly from Onsager's solution:

$$\frac{k_B T_c}{J} = \frac{2}{\ln(1 + \sqrt{2})} \approx 2.2691853, \quad (2)$$

or equivalently,

$$J\beta_c = J \frac{1}{k_B T_c} \approx 0.4406868, \quad (3)$$

where k_B is the Boltzmann constant. [1]

The order parameter is the magnetization per spin, defined as:

$$m = \frac{1}{N} \sum_{i=1}^N s_i, \quad (4)$$

where $N = L^2$ is the total number of spins in the lattice.

The zero-field susceptibility χ is defined thermodynamically as:

$$\chi = \left. \frac{\partial \langle m \rangle}{\partial h} \right|_{h=0} = \beta (\langle m^2 \rangle - \langle m \rangle^2). \quad (5)$$

By the fluctuation-dissipation theorem, this equals to

$$\chi = \beta N (\langle m^2 \rangle - \langle m \rangle^2). \quad (6)$$

In finite simulations at $h = 0$, the $\mathbb{Z}_2$ symmetry implies $\langle m \rangle = 0$ unless symmetry is explicitly broken. For thermodynamic estimates we use m in the fluctuation formula for χ , but for visualization we plot $\langle |m| \rangle$ in figures (in particular for the Wolff algorithm), since $|m|$ avoids sign cancellations and makes the ordered phase evident.

2.2 Monte Carlo Dynamics

2.2.1 Markov Chains and Master Equations

Monte Carlo updates define a Markov chain on the configuration space with a transition operator that has the Boltzmann distribution as its stationary measure. For relaxational models, such as the stochastic Ising model, the time-dependent behavior is described by a master equation

$$\frac{\partial P_n(t)}{\partial t} = - \sum_{n \neq m} [P_n(t) W_{n \rightarrow m} - P_m(t) W_{m \rightarrow n}], \quad (7)$$

where $P_n(t)$ is the probability of the system being in state n at time t , and $W_{n \rightarrow m}$ is the transition rate for $n \rightarrow m$.

The solution to the master equation is a sequence of states, but the time variable is a stochastic quantity which does not represent true time. A relaxation function $\phi(t)$ can be defined which describes time correlations *within equilibrium*

$$\phi_{MM}(t) = \frac{\langle M(0)M(t) \rangle - \langle M \rangle^2}{\langle M^2 \rangle - \langle M \rangle^2}. \quad (8)$$

When normalized in this way, the relaxation function is 1 at $t = 0$ and decays to zero as $t \rightarrow \infty$. It is important to remember that for a system in equilibrium any time in the sequence of states may be chosen as the ' $t = 0$ ' state. The asymptotic, long time behavior of the relaxation function is exponential, i.e.

$$\phi(t) \rightarrow e^{-t/\tau}, \quad (9)$$

where the correlation time τ diverges as T_c is approached.

2.2.2 Dynamic Critical Phenomena

This dynamic (relaxational) critical behavior can be expressed in terms of a power law as well,

$$\tau \propto \xi^z \propto \varepsilon^{-\nu z}, \quad (10)$$

where ξ is the (divergent) correlation length, $\varepsilon = |1 - T/T_c|$, and z is the dynamic critical exponent.

In a finite system of linear size L , the divergence of the correlation length at criticality is limited by the system size. Exactly at T_c , the correlation length therefore saturates at $\xi \sim L$. Inserting this finite-size cutoff into the critical slowing-down relation $\tau \propto \xi^z$ gives

$$\tau_c \sim L^z. \quad (11)$$

Thus, at the critical temperature, the relaxation time is no longer controlled by the reduced temperature ε , but instead by the system size itself. As L increases, correlations require more time to propagate across the entire system, leading to a relaxation time that scales as a power law in L with exponent z .

A second relaxation time, the integrated relaxation time, is defined by the integral of the relaxation function

$$\tau_{\text{int}} = \int_0^\infty \phi(t) dt. \quad (12)$$

This quantity is of particular importance for the determination of errors and is expected to diverge with the same dynamic exponent as the 'exponential' relaxation time, as both are governed by the same long-time behavior of $\phi(t)$.

- Explain that the Metropolis and Wolff algorithms represent distinct stochastic dynamics (local vs. cluster).

- Describe briefly the nature of updates each performs.

2.3 Dynamic Critical Phenomena

Here is where you derive the concept of dynamic critical exponents.

- Near T_c , the correlation length diverges as $\xi \sim |T - T_c|^{-\nu}$.

- Introduce time correlation functions, such as

$$C_X(t) = \langle X(0)X(t) \rangle - \langle X \rangle^2.$$

- Define the autocorrelation time τ_X via the exponential decay of $C_X(t)$:

$$C_X(t) \sim e^{-t/\tau_X}.$$

- State the scaling hypothesis:

$$\tau_X \sim \xi^z, \text{ where } z \text{ is the dynamic critical exponent.}$$

- Combining both gives the divergence near criticality:

$$\tau_X \sim |T - T_c|^{-z\nu}.$$

- Explain qualitatively that z depends on the update dynamics (hence on the algorithm).
This section should include equation derivations, reasoning, and physical interpretation.

3 Methodology

3.1 Practical Estimators and Extraction of z

To quantify critical slowing down we compute the autocorrelation function of an observable A (for instance the magnetization M , the energy or the susceptibility):

$$C_A(t) = \langle A_s A_{s+t} \rangle - \langle A \rangle^2,$$

and the normalized autocorrelation (autocorrelation coefficient)

$$\rho_A(t) = \frac{C_A(t)}{C_A(0)}.$$

A convenient scalar measure of the memory of the Markov chain is the *integrated autocorrelation time*

$$\tau_{\text{int},A} = \frac{1}{2} + \sum_{t=1}^{T_{\text{max}}} \rho_A(t),$$

where the sum is truncated at a cutoff T_{max} chosen e.g. by the window method (or until $\rho_A(t)$ has decayed to noise). The statistical error of a sample mean $\bar{A}$ computed from N recorded measurements is increased by autocorrelation; the effective sample size is $N_{\text{eff}} \approx N/(2\tau_{\text{int},A})$.

In practice we estimate $\tau_c(L)$ by the integrated autocorrelation time of the magnetization (or by the exponential autocorrelation time obtained from a fit of $\rho(t) \approx e^{-t/\tau_{\text{exp}}}$), and then fit $\log \tau_c$ versus $\log L$ to extract z :

$$\log \tau_c = z \log L + \text{const.}$$

3.2 Numerical Implementation

- Briefly describe the simulation parameters: lattice size, boundary conditions, temperature range around T_c , number of Monte Carlo steps.
- Explain data collection: measuring magnetization M , energy E , and recording time series.

3.3 Computation of Autocorrelation

- Explicit definition:

$$C_X(t) = \frac{\langle X_i X_{i+t} \rangle - \langle X \rangle^2}{\langle X^2 \rangle - \langle X \rangle^2}.$$

- Define integrated autocorrelation time or exponential autocorrelation time, and how you extract it (e.g., exponential fit, binning analysis).

3.4 Simulation Details

- Note equilibration time, number of independent runs, random number generator, etc.
- Briefly explain how T_c is estimated or chosen.

4 Results

4.1 Equilibrium Verification

Magnetization and energy vs. temperature.

4.2 Autocorrelation Behavior

- Show sample autocorrelation functions for Metropolis and Wolff updates.
- Discuss qualitative differences: slow decay for Metropolis near T_c , fast for Wolff.

4.3 Divergence near Criticality

- Plot autocorrelation time $\tau(T)$ vs. $|T - T_c|$.
- Fit $\tau \sim |T - T_c|^{-z\nu}$ or equivalently $\tau \sim L^z$ at T_c for finite-size scaling.
- Extract dynamic critical exponents $z_{\text{Metropolis}}$ and z_{Wolff}^{**} .

Expected behavior:

- Metropolis: $z \approx 2.1$
- Wolff: $z \approx 0-0.3$

4.4 Comparison of Algorithm Efficiency

- Compare CPU time per independent sample.
- Discuss how cluster algorithms reduce critical slowing down.

5 Discussion

- Interpret differences in dynamic exponents in terms of correlation spreading mechanisms.
- Discuss limits of each approach (finite-size effects, statistical errors).
- Relate results to universality class arguments.
- Possibly mention future work or extensions (e.g., Swendsen-Wang algorithm, higher dimensions).

6 Conclusion

- Summarize key findings: - Divergence of τ near T_c ,
- Distinct dynamic exponents for different algorithms,
- Algorithmic efficiency and importance for Monte Carlo near criticality.
- Mention physical significance: critical slowing down as an emergent dynamical effect in critical systems.

References

- [1] “Square lattice ising model.” Wikipedia, 2025. Accessed on: December 10, 2025.